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CONFORMATIONAL ANALYSIS OF 2-(α–D-LYXOFURANOSYL)MALEIMIDE

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ABSTRACT:

2-(α–D-lyxofuranosyl)maleimide is one of the analogues of the α anomer of the antitumour antibiotic showdomycin. Conformation of 2-(α–D-lyxofuranosyl)maleimide on the basis of potential energy calculation is reported. These calculations indicate that the drug has allowed conformation in the region of ω = 0˚ to 20˚, ω = 80˚ to 120˚ and at ω = 360˚. The region of ω = 140˚-340˚ is not allowed. The most feasible position for the drug to interact with receptor would be at minimum potential energy in the regions from ω = 0˚ to 20˚, ω = 80˚ to 120˚ and at ω = ω = 360˚.

INTRODUCTION

2-(α-D-lyxofuranosyl) maleimide is one of the α anomer of the antitumour antibiotic showdomycin. Showdomycin is a naturally occuring β-C-nucleoside with a maleimide group replacing a normal nucleobase (Suhadolnik, 1979). It is moderately active against gram-positive and gram-negative bacteria and shows some cytotoxic activity against HeLa cells. It is highly active against ascites tumours in rats, probably by means of inhibition of nucleic acid synthesis.

There is considerable current interest in nucleoside analogues, in large part because of antiviral activity shown by compounds such as 2′,3′-dideoxy-cytidine and 3′-azido-3′-deoxy-thymidine against human immunodeficiency virus (HIV) (Mitsuya & Broder, 1986; DeClereq, 1986). The latter compound has been established to act by blocking the action of the reverse transcriptase enzyme, RNA-dependent DNA polymerase (Furman *et al.,* 1986).

Nucleosides with the unusual α–configuration have also been reported to show this action, and some have potent experimental antitumour activity.

This paper reports on conformation of one of the analogues of showdomycin 2-(α-D-lyxofuranosyl) maleimide, which have an α – configuration at C1′. It crystallizes in monoclinic system $P2_1$ with unit cell dimensions a = 8.681 Å, b = 5.135 Å, c = 14.364 Å and β = 109.75º.

Synthetic studies on this compound have been published elsewhere (Kaye, Neidle & Reese, 1988 a,b).

The present work describes conformational analysis of 2-(α-D-lyxofuranosyl) maleimide similar to other drugs (Bano *et al*., 2003; Haleem *et al*., 1988; Akhtar *et al*., 1999 and Ahmed *et al*., 2003).

METHOD OF CALCULATION

The detailed method of calculation is given elsewhere (Haleem & Saify, 1986). Fig.1 shows projection of the molecule based upon the structural coordinates given by Stephen *et al* (1990). Bond length and bond angles have been calculated for 2-(α-D-lyxofuranosyl)maleimide. The contact distance and potential energies have been calculated for the following pairs (Kitaigorodsky, 1962).

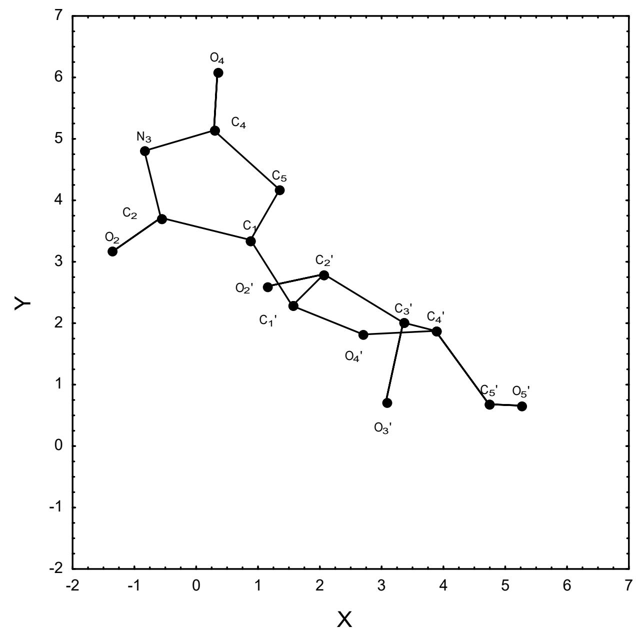

Fig.1: 001 projection of 2-(α–D-lyxofuranosyl)maleimide

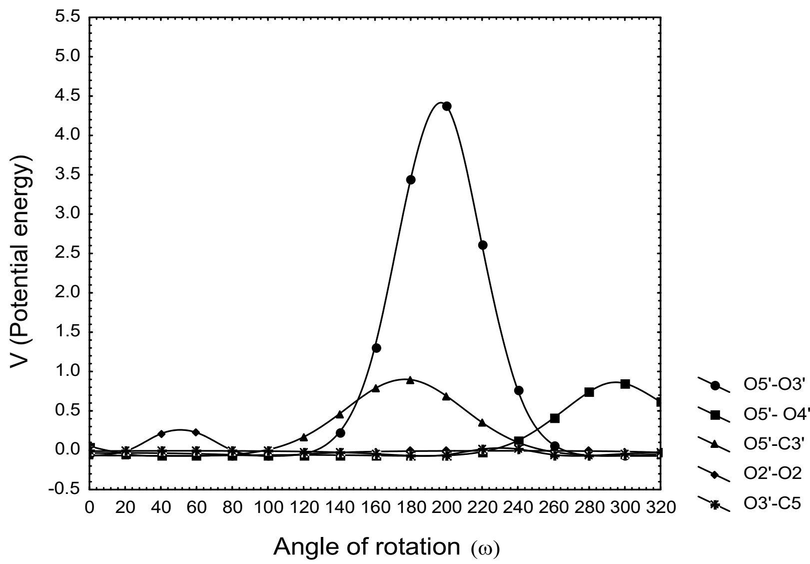

Fig. 2: Potential energy map for all the pairs.

O_5'--C_2', O_5'--O_2', O_5'--C_3', O_5'--O_3', O_5'--O_4'
O_2'--C_2, O_2'--O_2, O_2'--C_5,
O_3'--C_2, O_3'--O_2, O_3'--C_5.

In above pairs atoms O5′, O2′, O3′ are rotated from 0° to 360° about the bonds C_4'--C_5', C_1--C_1', C_5'--C_4' respectively. ω is the angle of rotation.

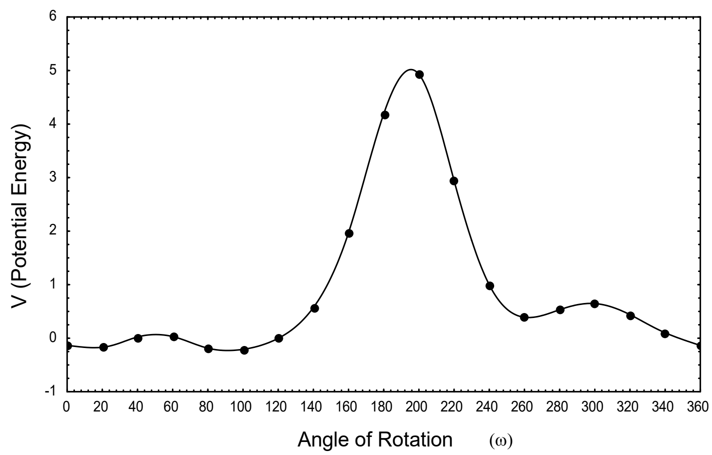

Fig. 3: Potential energy map for sum of all the pairs.

Table-1
Bond lengths of 2-(2-(α-D-lyxofuranosyl)maleimide

Atoms	Calculated bond lengths (Å)	Bond lengths given by Stephen *et al*. (Å)
C_1'-C_2'	1.544	1.545
C_1'-O_4'	1.420	1.419
C_1'-C_1	1.491	1.491
C_2'-O_2'	1.424	1.425
C_2'-C_3'	1.518	1.518
C_3'-O_3'	1.425	1.425
C_3'-C_4'	1.512	1.513
C_4'-C_5'	1.496	1.496
C_4'-O_4'	1.445	1.446
C_5'-O_5'	1.425	1.424
C_1-C_2	1.515	1.514
C_1-C_5	1.327	1.328
C_2-O_2	1.213	1.211
C_2-N_3	1.378	1.379
N_3-C_4	1.387	1.387
C_4-O_4	1.214	1.214
C_4-C_5	1.482	1.481

Several programs written in BASIC language and Pentium III computers were used through out the work.

Table-2
Bond angles of 2-(α-d-lyxofuranosyl)maleimide

Atoms	Calculated bond angles	Bond angles given by Stephen *et al*
C_2'-C_1'-O_4'	105.8	105.8
C_2'-C_1'-C_1	112.2	112.1
O_4'-C_1'-C_1	110.1	110.1
C_1'-C_2'-O_2'	113.3	113.3
C_1'-C_2'-C_3'	103.1	103.1
O_2'-C_2'-C_3'	112.2	112.1
C_2'-C_3'-O_3'	110.8	110.7
C_2'-C_3'-C_4'	102.4	102.3
O_3'-C_3'-C_4'	108.6	105.6
C_3'-C_4'-C_5'	115.5	115.4
C_3'-C_4'-O_4'	105.8	105.7
C_5'-C_4'-O_4'	109.7	109.6
C_4'-C_5'-O_5'	112.2	112.2
C_1'-O_4'-C_4'	110.8	110.8
C_1'-C_1-C_2	121.4	121.4
C_1'-C_1-C_5	130.9	130.9
C_2-C_1-C_5	107.7	107.6
C_1-C_2-O_2	127.4	127.4
C_1-C_2-N_3	106.4	106.3
O_2-C_2-N_3	126.2	126.2
C_2-N_3-C_4	110.1	110.1
N_3-C_4-O_4	123.9	123.8
N_3-C_4-C_5	107.0	107.0
O_4-C_4-C_5	129.1	129.1
C_1-C_5-C_4	108.9	108.8

RESULTS

Bond lengths and bond angles of 2-(α-D-lyxofuranosyl)maleimide are given in Table-1 and 2 respectively which are close to the bond lengths and bond angles calculated by Stephen *et al*. (1990).

Potential energies are calculated for several pairs but the serious type of interaction is found in the following pairs.

O_5'--C_3', O_5'--O_3', O_5'--O_4'
O_2'--O_2, O_3'--C_5

Potential energies of all the above pairs is shown in Fig.2.

O_5'--C_3'

The maximum potential energy is found to be 0.89673 Kcal / mole at ω = 180˚ and minimum potential energy is found to be –0.07211 Kcal / mole at ω = 300˚. The allowed regions are from ω = 0˚ to 80˚ and from ω = 260˚ to 360˚.

O_5'--O_3'

The maximum potential energy is found to be 4.3739 Kcal / mole at ω = 200˚ and minimum potential energy is found to be –0.07143767 Kcal / mole at ω = 100˚. The allowed regions are from ω = 0˚ to 120˚, ω = 280˚ to 360˚.

O_5'--O_4'

The maximum potential energy is found to be 0.8484 Kcal / mole at ω = 300˚ and minimum potential energy is found to be –0.071894 Kcal / mole at ω = 180˚. The allowed regions are from ω = 20˚ to 220˚.

O_2'--O_2

The maximum potential energy is found to be 0.22423 Kcal / mole at ω = 60˚ and minimum potential energy is found to be –0.07215 Kcal / mole at ω = 100˚. The allowed regions are from ω = 0˚ to 20˚ and from ω = 80˚ to 360˚.

O_3'--C_5

The maximum potential energy is found to be 0.01469 Kcal / mole at ω = 240˚ and minimum potential energy is found to be –0.06898 Kcal / mole at ω = 280˚. The allowed regions are from ω = 0˚ to 200˚ and from ω = 260˚ to 360˚.

Finally, sum of all the above pairs is taken and potential energies are calculated. Sum of potential energies of all the above pairs is shown in Fig. 3. The maximum potential energy is found to be 4.93114 Kcal / mole at ω=200˚. The allowed conformation for this drug is from ω = 0˚ to 20˚, ω = 80˚ to 120˚ and at ω = 360˚.

DISCUSSION

The main object of the present study is to add information on three dimensional structure of 2-(α–D-lyxofuranosyl)maleimide.

By these calculations it is possible to provide a much more detailed picture of active receptor site and conformation of the drug for the interaction with receptor. During the present work attempt has been made to propose the structure of 2-(α–D-lyxofuranosyl) maleimide in most active form. It was noticed that allowed region for 2-(α–D lyxofuranosyl) maleimide is from ω = 0° to 20°, ω = 80° to 120° and at ω = 360°. It is suggested that this drug will produce its pharmacological action in these regions.

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